

HW2 — Q2: Dogs vs. Cats Image Classification

Data Preprocessing

The pipeline handles all data loading and organization in three steps:

- Extraction: Training (25,000 images) and test (12,500 images) sets are unzipped from the Kaggle competition input directory.
- Organization: Images are copied into class-labelled subdirectories (cats/ and dogs/) required by Keras' `ImageDataGenerator.flow_from_directory()`.
- Normalization

Data Augmentation & Validation Strategy

Both models use an 80/20 train-validation split (20,000 train / 5,000 validation images)

Model Architectures

Model 1 - Custom CNN Baseline

Model 2 - ResNet50 Transfer Learning

Results

- CNN Baseline: Val Accuracy = 80.84%, Val Loss = 0.4113
- ResNet50: Val Accuracy = 98.50%, Val Loss = 0.0447
- Kaggle Public Leaderboard Score: 0.05899 (log loss)
- The ResNet50 model dramatically outperforms the CNN baseline, demonstrating the effectiveness of transfer learning — pretrained ImageNet features transfer strongly to animal classification tasks.